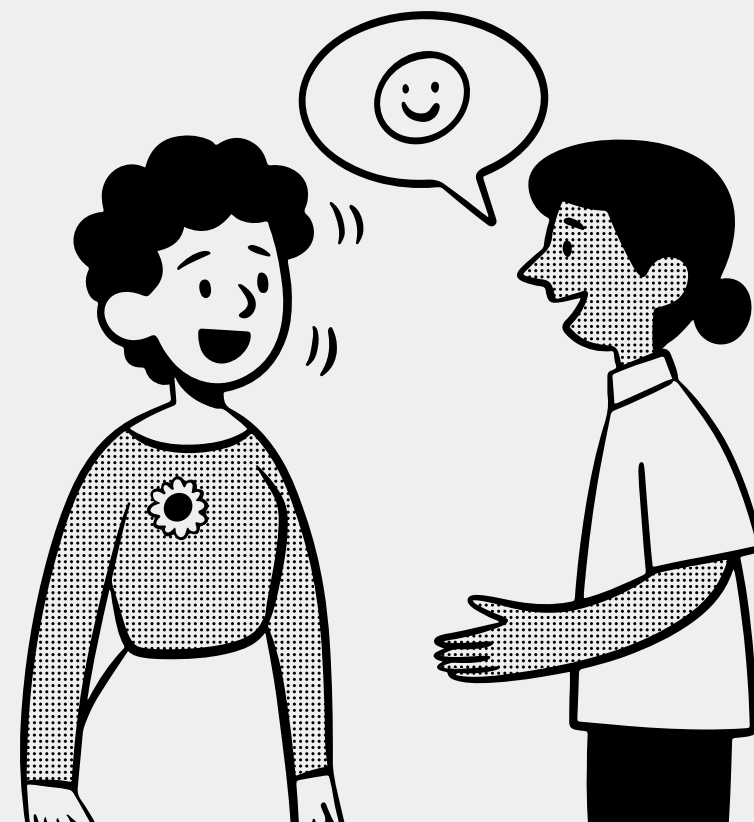


STRATASEARCH: ALGORITHMIC DOCUMENT SEARCH



*DSA-3 | GROUP 7 | REVIEW 4 — 100%
COMPLETE*



THE BLACK-BOX RETRIEVAL CRISIS

Opaque tools hide ranking mechanics and failure cases. Vector search can add GPU-heavy cost for exact questions.

StrataSearch delivers deterministic matching, formal Big-O, explainable selection, and lightweight browser or JVM execution.

WHY STRATASEARCH?





HYBRID
FALLBACK

CORE: pure Java 21 algorithms.
BACKEND: Spring Boot 3.3, async jobs, WebSockets.
FRONTEND: React 19 + TypeScript workbench.
CLIENT: dsa.ts | planner.ts | profiler.ts

ENGINEERING
ARCHITECTURE

ALGORITHMS & BIG-O GUARANTEES

KMP $O(n)$, space $O(m)$. Rabin-Karp avg $O(n+m)$, worst $O(nm)$. Z-Algorithm $O(n+m)$. Aho-Corasick $O(n + \text{matches})$. Suffix Array + LCP query $O(m \log n)$, space $O(n)$.

FORMAL COMPLEXITY MATRIX

5 CORE ALGORITHMS





VERIFIED
METRICS

Physical comparisons isolate theory from clock jitter. KMP and Z scale linearly with n . Aho-Corasick reaches $4.8\times$ throughput for 5+ patterns. Suffix Array breaks even after 4+ queries; later lookups stay below 2 ms.

EMPIRICAL
BENCHMARKING

QUERYPLANNER & COST MODEL

$\text{Cost} = \text{Wsize} \cdot \text{Csize} + \text{Wpattern} \cdot \text{Cpattern} + \text{Wrepeat} \cdot \text{Crepeat} + \text{Wentropy} \cdot \text{Centropy} + \text{Wquery} \cdot \text{Cquery} + \text{Bias}$. Weights: .35/.25/.20/.20/.10. The planner compares algorithms, explains trade-offs, and reports confidence.

EXPLAINABLE

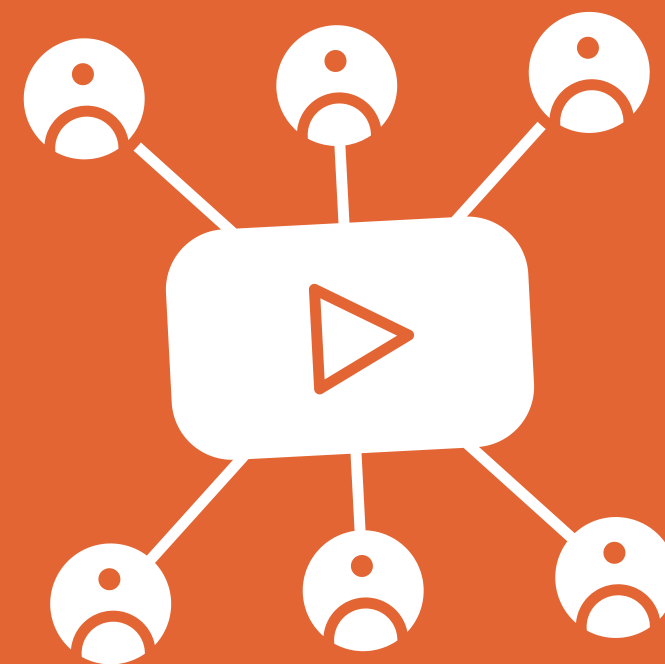


CORPUS PROFILING & SUSTAINABILITY

LOW-CARBON RETRIEVAL

Entropy measures information density; TTR measures vocabulary richness; bigrams reveal repetition. Profiles classify LARGE_SCALE, HIGH_ENTROPY, or INDEX_FRIENDLY. Exact search avoids GPU/vector overhead; browser RAM stays below 50 MB.





100% COMPLETE
BALAN

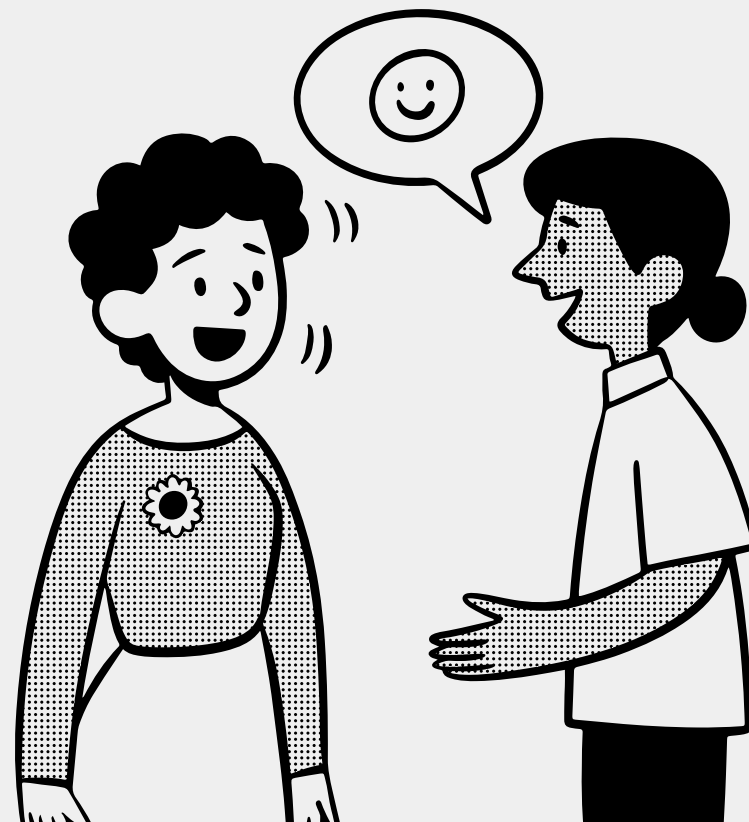
SOCIETY: open-source analysis, citation provenance, and visual algorithm stepping.

TEAM: Sasank — planner, KMP, Aho-Corasick, deployment. Rithvik — suffix array, Rabin-Karp, backend. Arnavi — Z-Algorithm, visualizer, telemetry.

ROADMAP: Reviews 1–3 delivered; Review 4 completes profiling, hybrid execution, DOCX, CI/CD, and tests.

CED
**UTILITY, TEAM
& DELIVERY**

SUMMARY & LIVE DEMO



LIVE DEMO & Q&A

